Reproducible Omics Evidence Audit

Leakage diagnostics, stability, and transportability for high-dimensional classifiers

1 PUBLIC COHORTS

Breast cancer pCR/RD

N = 306
GSE25055 discovery

N = 182
GSE25065 same-family

N = 253
GSE41998 cross-platform

GSE41998 endpoint audit: 69 Yes, 184 No; 20 literal zeros and 6 missing excluded.

No ambiguous response recoding

2 WORKFLOW CONTRAST

Naive workflow

Global supervised feature selection before repeated CV; fixed C = 1.

Guarded workflow

Feature selection, scaling, weighting, and C tuning confined to training partitions.

0.783
naive AUROC

0.703
guarded AUROC

Delta AUROC = +0.080
95% CI +0.008 to +0.151

Descriptive whole-workflow contrast

3 AUDIT DIAGNOSTICS

0.878
naive shuffled-label null

0.492
guarded null

0.513
Nogueira stability

28 / 30
positive AUROC contrasts

Frozen external AUROC

0.563
GSE25065

0.692
GSE41998

Operating-point behavior changed by cohort and scaling even when AUROC was similar.

Uncertainty + null + stability + external checks

4 INTERPRETATION

1. The naive null diagnoses an invalid workflow.
2. Leakage control does not guarantee transportability.
3. Feature recurrence is an audit output, not a biomarker.
4. The unmatched workflow contrast is not a causal leakage effect.

Reusable credibility audit

Not clinical biomarker discovery

ROEA asks whether a strong result is credible, stable, and transportable

Breast cancer pCR/RD is the stress test; the contribution is the evidence-audit framework.